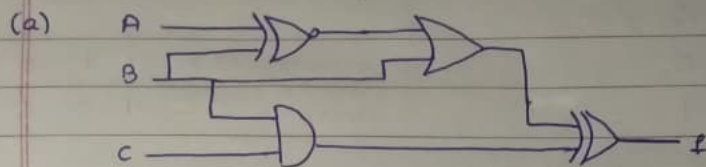


1) Derive the output expression 'f' for the below circuits and draw the simplified circuits.



→ From the circuit,

• First Gate = XOR

$$(A \oplus B)' = AB + A'B'$$

• Second Gate = OR

$$(A \oplus B)' + B$$

• Bottom Gate = AND

$$BC$$

• Final Gate = XOR

$$\text{output expression : } f = ((A \oplus B)' + B) \oplus (BC)$$

Simplify:

$$AB + B = B(A + 1) = B$$

$$(A \oplus B)' = AB + A'B'$$

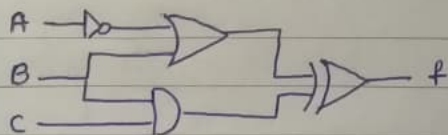
$$B + A'B' = (A' + B)(B + B')$$

$$AB + A'B' + B = B + A'B' = B + A'$$

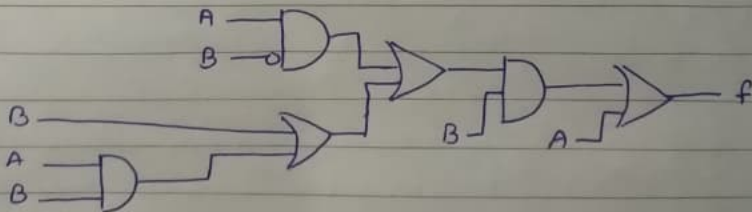
$$= A' + B$$

∴ simplified Expression:

$$f = (A' + B) \oplus BC$$



(b)



→ from the circuit,

- Top AND: AB'
- Bottom AND: AB
- First OR: $B + AB = B$
- Second OR: $AB' + B = A + B$
- AND with B:

$$(A+B)B = AB + BB = AB + B = B(A+1) = B$$

- last OR:

$$B + A = A + B$$

output expression: $f = [(AB') + (B + AB)]B + A$

simplified expression:

$$f = A + B$$



(c)



→ from the circuit

- NOT on A: $\bar{A}$
- OR: $\bar{A} + B$
- NOT: $\overline{(\bar{A} + B)} = \bar{\bar{A}} \bar{B} = A \bar{B}$
- AND: $A \bar{B} B = 0$
- NOT: $\bar{0} = 1$
- AND: $1 \cdot C = C$
- OR: $B + C$

$$\overline{(\bar{A} + B)}$$

$$\bar{\bar{A}} \bar{B}$$

output expression: $f = [\overline{((\bar{A} + B)B)}C] + B$

$$f = [((A' + B)'B)'C] + B$$

simplified expression:

$$f = B + C$$



2] Minimize the following Boolean Expressions using Boolean postulates

(a) $\overline{A}\overline{C}\overline{D}E + \overline{A}\overline{B}\overline{D} + ABCE + ABD$

→ $\overline{A}\overline{D}(CE + \overline{B}) + AB(CE + D)$

(b) $\overline{A}\overline{B}\overline{C} + ABD + \overline{A}\overline{C} + \overline{A}\overline{C}\overline{D} + A\overline{C}D + A\overline{B}\overline{C}$

→ $\overline{A}\overline{C} + \overline{A}\overline{C}\overline{D} = \overline{A}\overline{C}$ absorption law: $X + XY = X$

$\overline{A}\overline{B}\overline{C} + ABD + \overline{A}\overline{C} + A\overline{C}\overline{D} + A\overline{B}\overline{C}$

$\overline{A}\overline{B}\overline{C} + A\overline{B}\overline{C} = \overline{B}\overline{C}(\overline{A} + A) = \overline{B}\overline{C}$

$\overline{B}\overline{C} + ABD + \overline{A}\overline{C} + A\overline{C}\overline{D}$

$ABD + \overline{A}\overline{C} + \overline{C}(\overline{B} + AD)$

(c) $\overline{A}\overline{C} + BC + A\overline{B} + \overline{A}BC + AC\overline{D} + \overline{B}\overline{C}\overline{D}$

→ $\overline{A}\overline{B}C + BC = BC$ absorption law: $X + XY = X$

$\overline{A}\overline{C} + BC + A\overline{B} + AC\overline{D} + \overline{B}\overline{C}\overline{D}$

$\overline{C}[\overline{A} + \overline{B}\overline{D}] + BC + A\overline{B} + AC\overline{D}$

$A\overline{B} + \overline{C}[\overline{A} + \overline{B}\overline{D}] + C[\overline{B} + A\overline{D}]$

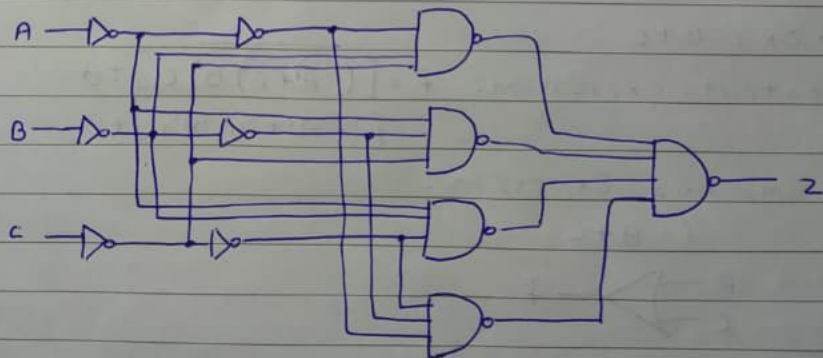
(d) $\overline{x}_1\overline{x}_3\overline{x}_4 + x_3x_4 + \overline{x}_1\overline{x}_2x_4 + \overline{x}_1x_2\overline{x}_3x_4 + x_2$

$x_2 + \overline{x}_1x_2\overline{x}_3x_4 = x_2$ absorption law: $X + XY = X$

$\overline{x}_1\overline{x}_3\overline{x}_4 + x_3x_4 + \overline{x}_1\overline{x}_2x_4 + x_2$

$\overline{x}_1[\overline{x}_3\overline{x}_4 + \overline{x}_2x_4] + x_3x_4 + x_2$

3] which single gate can represent the functionality of given circuit?



→ The first 4 Nand gates give 4 minterms corresponding to the cases where an odd number of inputs are 1.

$$\cdot \bar{A}, \bar{B}, C \quad \cdot \bar{A}, B, \bar{C} \quad \cdot A, \bar{B}, \bar{C} \quad \cdot A, B, C$$

The final Nand combines these outputs giving

$$Z = \overline{\bar{P}_1 \cdot \bar{P}_2 \cdot \bar{P}_3 \cdot \bar{P}_4} = P_1 + P_2 + P_3 + P_4$$

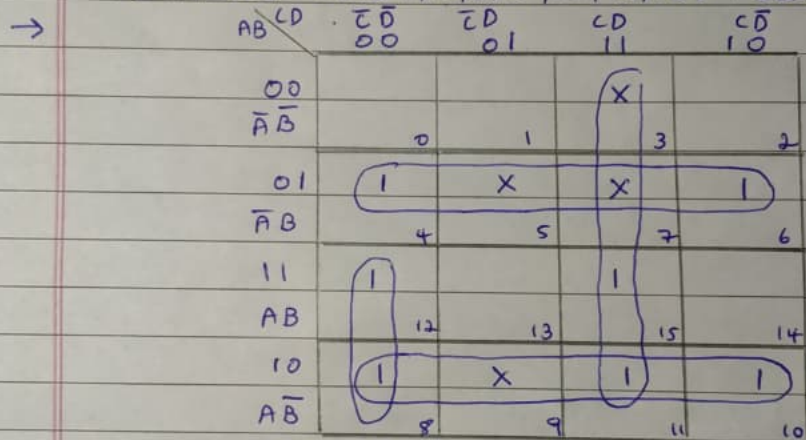
$$Z = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

This is exactly the boolean expression for a 3-input Exclusive-OR (XOR) gate:

$$Z = A \oplus B \oplus C$$

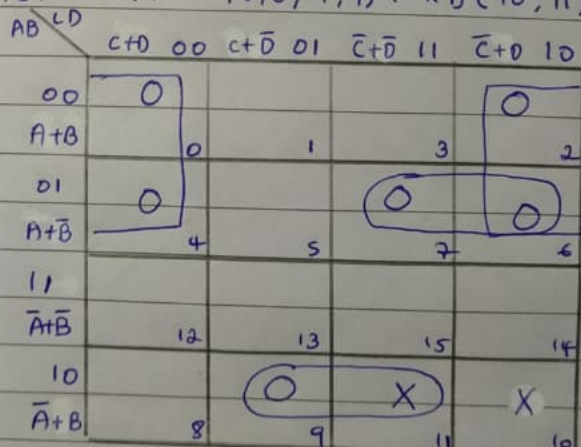
4) Minimize the following Boolean Expressions using K-Map.

(a) $f(a, b, c, d) = \sum m(4, 6, 8, 10, 11, 12, 15) + \sum d(3, 5, 7, 9)$



$$\therefore f = \bar{A}\bar{B} + \bar{A}B + CD + A\bar{C}\bar{D}$$

(b) $f(a, b, c, d) = \pi M(0, 2, 4, 6, 7, 9) + \pi D(10, 11)$



$$\therefore f = (A+B)(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{D})$$